

Generalised Linear Models

Logistic regression

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Q: Survival of passengers on the Titanic ~ Class

Read `titanic_long.csv` dataset and fit linear model (survival ~ class).

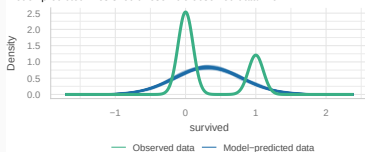
	class	age	sex	survived
1	first	adult	male	1
2	first	adult	male	1
3	first	adult	male	1
4	first	adult	male	1
5	first	adult	male	1
6	first	adult	male	1

Let's check linear model:

```
m5 <- lm(survived ~ class, data = titanic)
library('easystats')
check_model(m5)
```

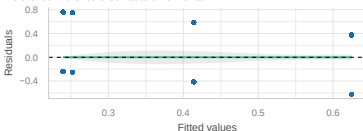
Posterior Predictive Check

Model-predicted lines should resemble observed data line



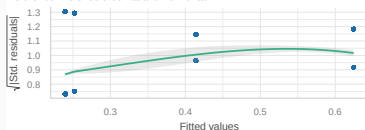
Linearity

Reference line should be flat and horizontal



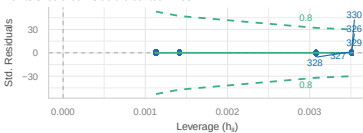
Homogeneity of Variance

Reference line should be flat and horizontal



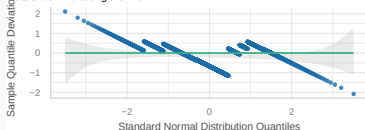
Influential Observations

Points should be inside the contour lines

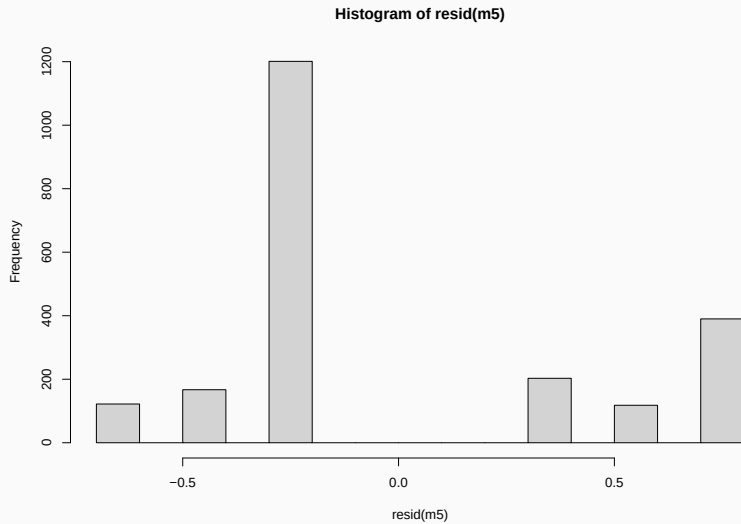


Normality of Residuals

Points should fall along the line



Weird residuals!



What if your residuals are clearly non-normal
or variance not constant (heteroscedasticity)?

- Binary variables (0/1)

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- Counts (0, 1, 2, 3, ...)

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- Categories ('small', 'medium', 'large'...)

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- Binary variables (0/1)
- Counts (0, 1, 2, 3, ...)
- Categories ('small', 'medium', 'large'...)
- **Generalised Linear Models to the rescue!**

1. Response variable - distribution family

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- Bernoulli - Binomial

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- etc

1. **Response variable** - distribution family

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- etc

2. **Predictors** (continuous or categorical)

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3. Link function

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- Gaussian: identity

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- Gaussian: identity
- Binomial: logit, probit

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- Binomial: logit, probit
- Poisson: log...

1. Response variable - distribution family

- Bernoulli - Binomial
- Poisson
- Gamma
- etc

2. Predictors (continuous or categorical)

3. Link function

- Gaussian: identity
- Binomial: logit, probit
- Poisson: log...
- See **family**.

The modelling process

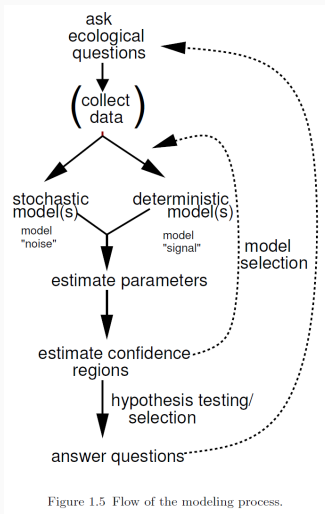


Figure 1.5 Flow of the modeling process.

Bernoulli - Binomial distribution (Logistic regression)

Response variable: **Yes/No** (e.g. survival, sex, presence/absence)

Canonical link function: **logit** (*log odds*), but others possible (see **family**)

$$\text{logit}(p) = \log\left(\frac{p}{1-p}\right)$$

Then

$$\text{logit}(P(\text{alive})) = a + bx$$

$$P(\text{alive}) = \text{invlogit}(a + bx) = \frac{e^{a+bx}}{1 + e^{a+bx}}$$

Where is the variance?

In a Gaussian GLM

$$y \sim \text{Normal}(\mu, \sigma)$$

In a Binomial GLM

$$y \sim \text{Binomial}(n, p)$$

n = number of trials

p = probability of success

$$\text{Var}(y) = np(1 - p)$$

(maximum variance when **p** around 0.5)

Back to survival of Titanic passengers

How many survived in each class?

```
table(titanic$class, titanic$survived)
```

	0	1
crew	673	212
first	122	203
second	167	118
third	528	178

How many survived in each class? (*dplyr*)

```
titanic |>  
  group_by(class, survived) |>  
  summarise(count = n())
```

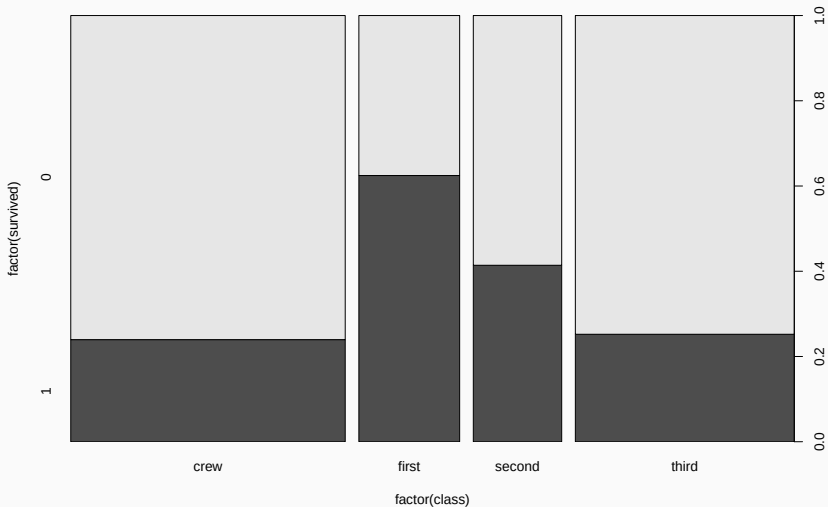
```
# A tibble: 8 x 3
```

```
# Groups:   class [4]
```

	class	survived	count
	<chr>	<int>	<int>
1	crew	0	673
2	crew	1	212
3	first	0	122
4	first	1	203
5	second	0	167
6	second	1	118
7	third	0	528
8	third	1	178

Data visualisation (mosaic plot)

```
plot(factor(survived) ~ factor(class), data = titanic)
```



```
tit.glm <- glm(survived ~ class,  
               data = titanic,  
               family = binomial)
```

which corresponds to

$$\text{logit}(P(\text{survival})_i) = a + b \cdot \text{class}_i$$

$$\text{logit}(P(\text{survival})_i) = a + b_{\text{first}} + c_{\text{second}} + d_{\text{third}}$$

Interpreting binomial GLM

```
tit.glm <- glm(survived ~ class, data = titanic, family = binomial)
```

Call:

```
glm(formula = survived ~ class, family = binomial, data = titanic)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-1.15516	0.07876	-14.667	< 2e-16 ***
classfirst	1.66434	0.13902	11.972	< 2e-16 ***
classecond	0.80785	0.14375	5.620	1.91e-08 ***
classtthird	0.06785	0.11711	0.579	0.562

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 2769.5 on 2200 degrees of freedom
Residual deviance: 2588.6 on 2197 degrees of freedom
AIC: 2596.6

Number of Fisher Scoring iterations: 4

Interpreting logistic regression output (easystats)

```
library('easystats') # 'modelbased' pkg  
estimate_means(tit.glm)
```

Estimated Marginal Means

class	Probability	95% CI
crew	0.24	[0.21, 0.27]
first	0.62	[0.57, 0.68]
second	0.41	[0.36, 0.47]
third	0.25	[0.22, 0.29]

Variable predicted: survived

Predictors modulated: class

Predictions are on the response-scale.

Analysing differences among factor levels (class)

```
estimate_contrasts(tit.glm)
```

Marginal Contrasts Analysis

Level1	Level2	Difference	SE	95% CI	z	p

first	crew	0.39	0.03	[0.33, 0.44]	12.65	< .001
second	crew	0.17	0.03	[0.11, 0.24]	5.37	< .001
third	crew	0.01	0.02	[-0.03, 0.06]	0.58	0.563
second	first	-0.21	0.04	[-0.29, -0.13]	-5.31	< .001
third	first	-0.37	0.03	[-0.43, -0.31]	-11.85	< .001
third	second	-0.16	0.03	[-0.23, -0.10]	-4.84	< .001

Variable predicted: survived

Predictors contrasted: class

p-values are uncorrected.

Contrasts are on the response-scale.

```
library('easystats')    # 'performance' pkg  
r2(tit.glm)
```

```
# R2 for Logistic Regression  
Tjur's R2: 0.087
```

But there are caveats (e.g. see [here](#) and [here](#))

```
kable(xtable::xtable(tit.glm), digits = 2)
```

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-1.16	0.08	-14.67	0.00
classfirst	1.66	0.14	11.97	0.00
classecond	0.81	0.14	5.62	0.00
classtthird	0.07	0.12	0.58	0.56

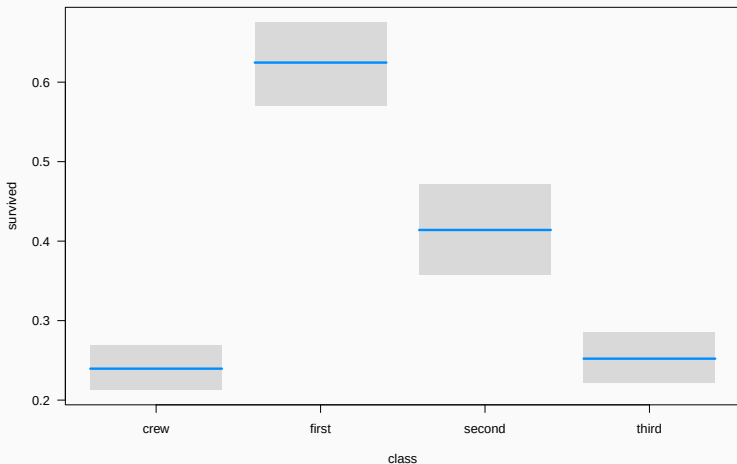
Presenting model results

```
library('modelsummary')  
modelsummary(tit.glm, output = 'markdown')
```

	(1)
(Intercept)	-1.155 (0.079)
classfirst	1.664 (0.139)
classecond	0.808 (0.144)
classtthird	0.068 (0.117)
Num.Obs.	2201
AIC	2596.6
BIC	2619.3
Log.Lik.	-1294.278
F	57.742

Visualising model: visreg package

```
visreg(tit.glm, scale = 'response', rug = FALSE)
```



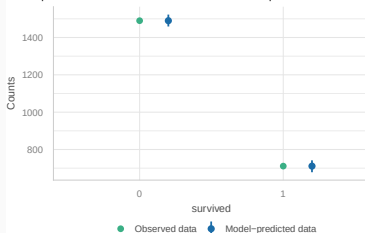
Model checking

check_model(easystats)

check_model(tit.glm)

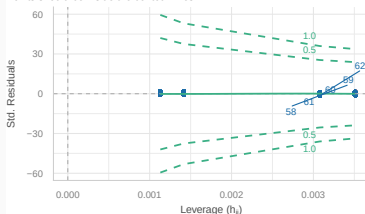
Posterior Predictive Check

Model-predicted intervals should include observed data points



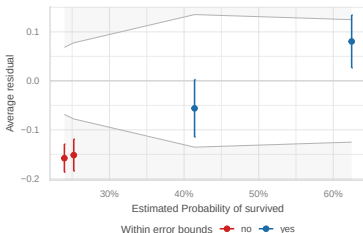
Influential Observations

Points should be inside the contour lines



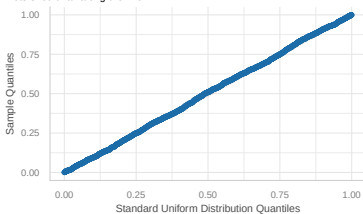
Binned Residuals

Points should be within error bounds



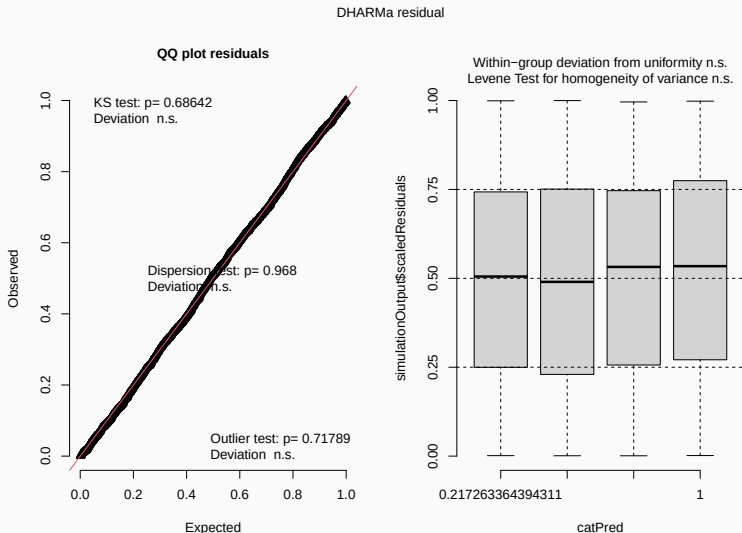
Distribution of Quantile Residuals

Dots should fall along the line



Residual diagnostics with DHARMa

```
library('DHARMa')  
simulateResiduals(tit.glm, plot = TRUE)
```



The goal is not to test whether the model's assumptions are 'true', because all models are false.

Rather, the goal is to assess exactly **how the model fails to describe the data**, as a path towards **model comprehension, revision, and improvement**.

Richard McElreath. *Statistical Rethinking*

1. Visualise data

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2. Fit model: `glm`. Don't forget to specify `family`!

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3. Examine model: `summary`

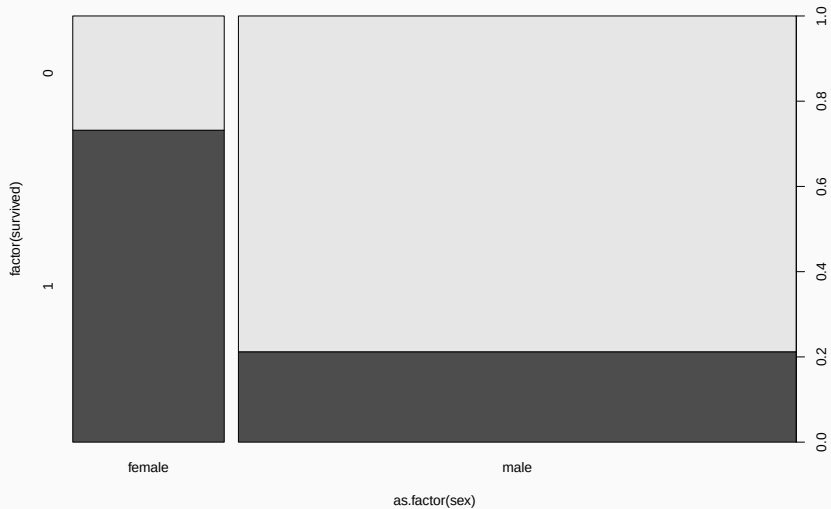
1. Visualise data
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4. Back-transform parameters from *logit* into probability scale
(`estimate_means`)

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3. Examine model: `summary`
4. Back-transform parameters from *logit* into probability scale
(`estimate_means`)
5. Plot model: `visreg`, ...
6. Check model: `check_model`, `DHARMA::simulateResiduals`,
`calibration_plot`

Q: Did men have higher survival
than women?

First, visualise data



Call:

```
glm(formula = survived ~ sex, family = binomial, data = titanic)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	1.0044	0.1041	9.645	<2e-16 ***
sexmale	-2.3172	0.1196	-19.376	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 2769.5 on 2200 degrees of freedom
Residual deviance: 2335.0 on 2199 degrees of freedom
AIC: 2339

Number of Fisher Scoring iterations: 4

Model interpretation

```
estimate_means(tit.sex)
```

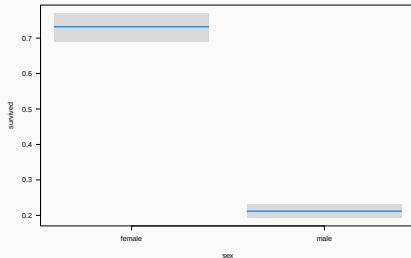
Estimated Marginal Means

sex	Probability	95% CI
female	0.73	[0.69, 0.77]
male	0.21	[0.19, 0.23]

Variable predicted: survived

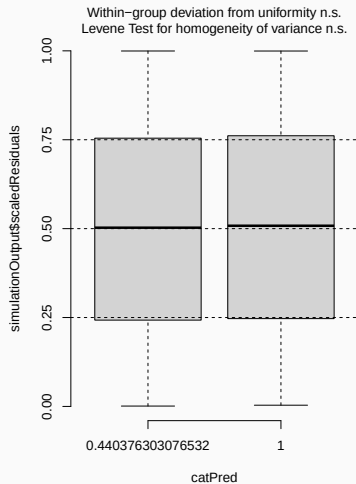
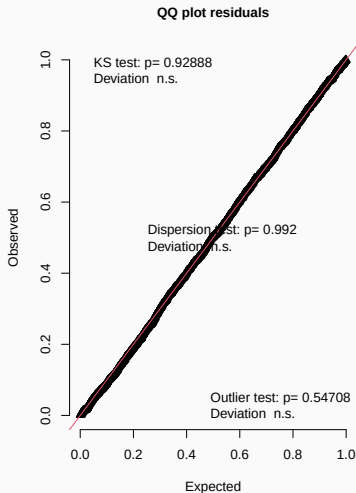
Predictors modulated: sex

Predictions are on the response-scale.




```
simulateResiduals(tit.sex, plot = TRUE)
```

DHARMA residual



Q: Did women have higher survival because they travelled more in first class?

Let's look at the data

```
table(titanic$class, titanic$survived, titanic$sex)
```

```
, , = female
```

	0	1
crew	3	20
first	4	141
second	13	93
third	106	90

```
, , = male
```

	0	1
crew	670	192
first	118	62
second	154	25
third	422	88

Fit additive model with both factors

Call:

```
glm(formula = survived ~ class + sex, family = binomial, data = titanic)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)	
(Intercept)	1.18740	0.15747	7.541	4.68e-14	***
classfirst	0.88081	0.15697	5.611	2.01e-08	***
classecond	-0.07178	0.17093	-0.420	0.675	
classthird	-0.77742	0.14231	-5.463	4.69e-08	***
sexmale	-2.42133	0.13909	-17.408	< 2e-16	***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

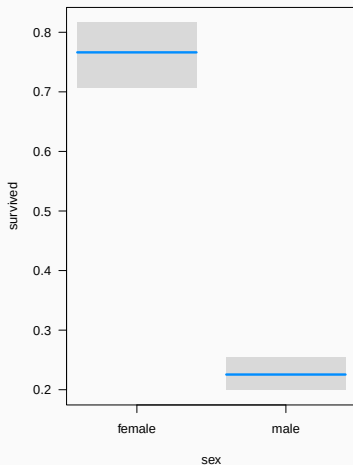
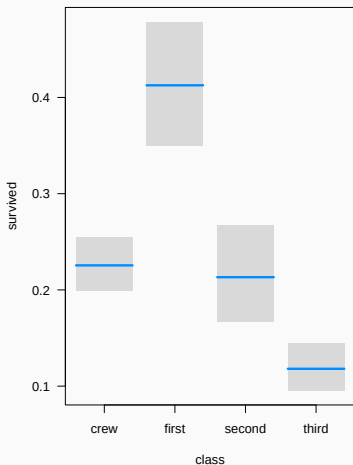
(Dispersion parameter for binomial family taken to be 1)

Null deviance: 2769.5 on 2200 degrees of freedom
Residual deviance: 2228.9 on 2196 degrees of freedom
AIC: 2238.9

Number of Fisher Scoring iterations: 4

Plot additive model

```
visreg(tit.sex.class.add, scale = 'response', rug = FALSE)
```



null device

Fit model with the interaction of both factors

Call:

```
glm(formula = survived ~ class * sex, family = binomial, data = titanic)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)	
(Intercept)	1.89712	0.61914	3.064	0.00218	**
classfirst	1.66535	0.80026	2.081	0.03743	*
classecond	0.07053	0.68630	0.103	0.91815	
classtthird	-2.06075	0.63551	-3.243	0.00118	**
sexmale	-3.14690	0.62453	-5.039	4.68e-07	***
classfirst:sexmale	-1.05911	0.81959	-1.292	0.19627	
classecond:sexmale	-0.63882	0.72402	-0.882	0.37760	
classtthird:sexmale	1.74286	0.65139	2.676	0.00746	**

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 2769.5 on 2200 degrees of freedom

Residual deviance: 2163.7 on 2193 degrees of freedom

AIC: 2179.7

Women had higher survival than men, even within the same class

```
estimate_means(tit.sex.class.int)
```

Estimated Marginal Means

class	sex	Probability	95% CI

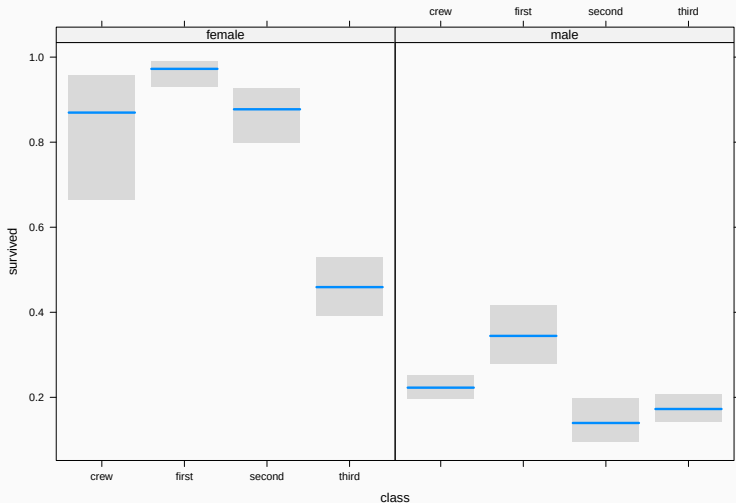
crew	female	0.87	[0.66, 0.96]
first	female	0.97	[0.93, 0.99]
second	female	0.88	[0.80, 0.93]
third	female	0.46	[0.39, 0.53]
crew	male	0.22	[0.20, 0.25]
first	male	0.34	[0.28, 0.42]
second	male	0.14	[0.10, 0.20]
third	male	0.17	[0.14, 0.21]

Variable predicted: survived

Predictors modulated: class, sex

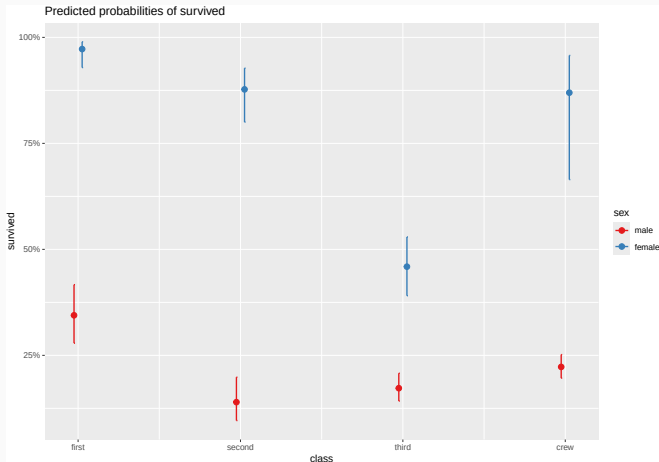
Women had higher survival than men, even within the same class

```
visreg(tit.sex.class.int, by = 'sex', xvar = 'class', scale = 'response', rug = FALSE)
```



Visualising model (sjPlot)

```
library('sjPlot')  
plot_model(tit.sex.class.int, type = 'int')
```



Comparing models

```
library('easystats') # 'performance' pkg
compare_performance(tit.sex.class.add, tit.sex.class.int)
```

Comparison of Model Performance Indices

Name	Model	AIC (weights)	AICc (weights)	BIC (weights)
tit.sex.class.add	glm	2238.9 (<.001)	2238.9 (<.001)	2267.4 (<.001)
tit.sex.class.int	glm	2179.7 (>.999)	2179.8 (>.999)	2225.3 (>.999)

Name	Tjur's R2	RMSE	Sigma	Log_loss	Score_log
tit.sex.class.add	0.248	0.405	1	0.506	-Inf
tit.sex.class.int	0.271	0.399	1	0.492	-Inf

Name	Score_spherical	PCP
tit.sex.class.add	0.004	0.671
tit.sex.class.int	0.002	0.681

Logistic regression for proportion data

Why not modelling proportion?

- $1/4 = 0.25$

Why not modelling proportion?

- $1/4 = 0.25$
- $250/1000 = 0.25$

Why not modelling proportion?

- $1/4 = 0.25$
- $250/1000 = 0.25$
- Sample size information is lost!

Read Titanic data in different format

Read `titanic_prop.csv` data.

	X	Class	Sex	Age	No	Yes
1	1	1st	Female	Adult	4	140
2	2	1st	Female	Child	0	1
3	3	1st	Male	Adult	118	57
4	4	1st	Male	Child	0	5
5	5	2nd	Female	Adult	13	80
6	6	2nd	Female	Child	0	13

These are the same data, but summarized (see `Freq` variable).

Use `cbind(n.success, n.failures)` as response

```
prop.glm <- glm(cbind(Yes, No) ~ Class, data = tit.prop, family = binomial)
```

Call:

```
glm(formula = cbind(Yes, No) ~ Class, family = binomial, data = tit.prop)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	0.5092	0.1146	4.445	8.79e-06 ***
Class2nd	-0.8565	0.1661	-5.157	2.51e-07 ***
Class3rd	-1.5965	0.1436	-11.114	< 2e-16 ***
ClassCrew	-1.6643	0.1390	-11.972	< 2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 671.96 on 13 degrees of freedom
Residual deviance: 491.06 on 10 degrees of freedom
AIC: 545.68

Number of Fisher Scoring iterations: 4

Survival probability by class

```
estimate_means(prop.glm)
```

Estimated Marginal Means

Class	Probability	95% CI
1st	0.62	[0.57, 0.68]
2nd	0.41	[0.36, 0.47]
3rd	0.25	[0.22, 0.29]
Crew	0.24	[0.21, 0.27]

Variable predicted: cbind(Yes, No)

Predictors modulated: Class

Predictors averaged: Yes (44), No (93)

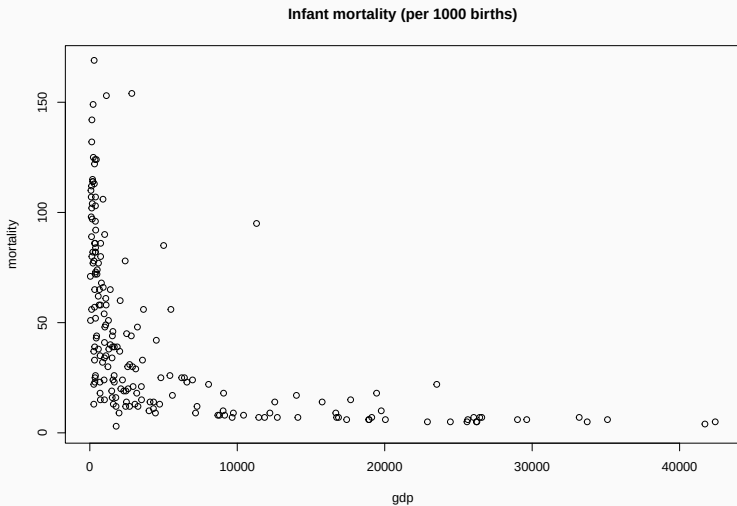
Predictions are on the response-scale.

Logistic regression with continuous predictors

Example dataset: GDP and infant mortality

Read UN_GDP_infantmortality.csv.

country	mortality	gdp
Length:207	Min. : 2.00	Min. : 36
Class :character	1st Qu.: 12.00	1st Qu.: 442
Mode :character	Median : 30.00	Median : 1779
	Mean : 43.48	Mean : 6262
	3rd Qu.: 66.00	3rd Qu.: 7272
	Max. :169.00	Max. :42416
	NA's :6	NA's :10



Fit model

```
gdp.glm <- glm(cbind(mortality, 1000 - mortality) ~ gdp,  
               data = gdp, family = binomial)
```

Call:

```
glm(formula = cbind(mortality, 1000 - mortality) ~ gdp, family = binomial,  
     data = gdp)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-2.657e+00	1.311e-02	-202.76	<2e-16 ***
gdp	-1.279e-04	3.458e-06	-36.98	<2e-16 ***

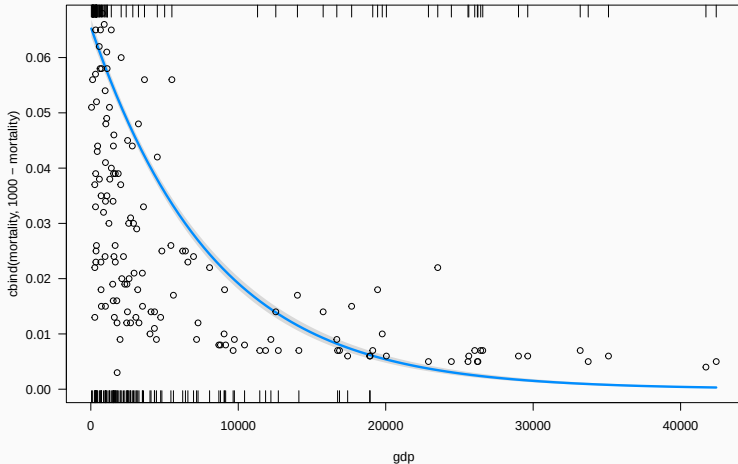
Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 6430.2 on 192 degrees of freedom
Residual deviance: 3530.2 on 191 degrees of freedom
(14 observations deleted due to missingness)
AIC: 4525.8

Plot model using visreg:

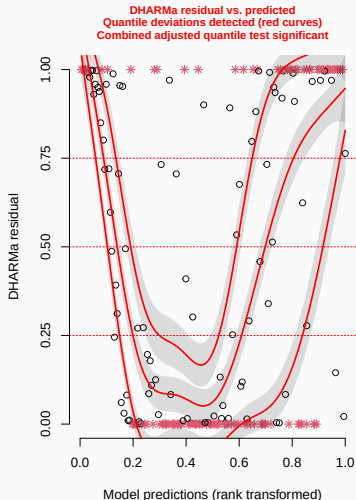
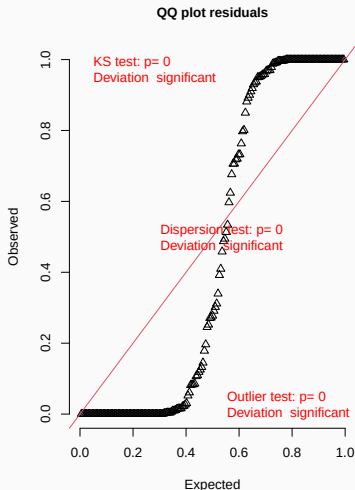
```
visreg(gdp.glm, scale = 'response')  
points(mortality/1000 ~ gdp, data = gdp)
```



Residuals diagnostics with DHARMA

```
simulateResiduals(gdp.glm, plot = TRUE)
```

DHARMA residual



Overdispersion

Overdispersion:

more variation in the data than assumed by statistical model

$$\text{Var}(y) = np(1 - p)$$

Testing for overdispersion (DHARMA)

```
simres <- simulateResiduals(gdp.glm, refit = TRUE)  
testDispersion(simres, plot = FALSE)
```

DHARMA nonparametric dispersion test via mean deviance residuals
vs. simulated-refitted

```
data:  simres  
dispersion = 21, p-value < 2.2e-16  
alternative hypothesis: two.sided
```

```
check_overdispersion(gdp.glm)
```

```
# Overdispersion test
```

```
dispersion ratio = 2.933
```

```
p-value = < 0.001
```

`quasibinomial` allows us to model overdispersed binomial data

Overdispersion in logistic regression with proportion data

```
gdp.overdisp <- glm(cbind(mortality, 1000 - mortality) ~ gdp,  
                    data = gdp, family = quasibinomial)
```

Call:

```
glm(formula = cbind(mortality, 1000 - mortality) ~ gdp, family = quasibinomial,  
    data = gdp)
```

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-2.657e+00	5.977e-02	-44.465	< 2e-16 ***
gdp	-1.279e-04	1.577e-05	-8.111	5.96e-14 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for quasibinomial family taken to be 20.7947)

Null deviance: 6430.2 on 192 degrees of freedom
Residual deviance: 3530.2 on 191 degrees of freedom
(14 observations deleted due to missingness)
AIC: NA

Mean estimates do not change after accounting for overdispersion

But standard errors (uncertainty) do!

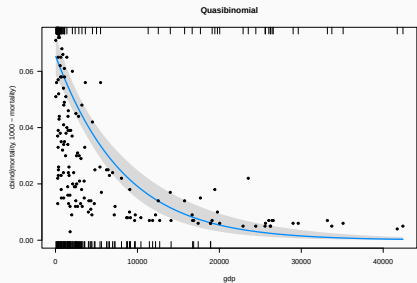
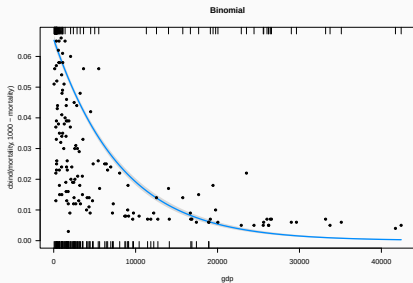
```
parameters(gdp.overdisp)
```

Parameter	Log-Odds	SE	95% CI	t(191)	p
(Intercept)	-2.66	0.06	[-2.78, -2.54]	-44.46	< .001
gdp	-1.28e-04	1.58e-05	[0.00, 0.00]	-8.11	< .001

```
parameters(gdp.glm)
```

Parameter	Log-Odds	SE	95% CI	z	p
(Intercept)	-2.66	0.01	[-2.68, -2.63]	-202.76	< .001
gdp	-1.28e-04	3.46e-06	[0.00, 0.00]	-36.99	< .001

But standard errors (uncertainty) do!



Moth predation

The industrial revolution and evolution of dark morphs



```
moth <- read.csv('data/moth.csv')
```

	MORPH	DISTANCE	PLACED	REMOVED
1	light	0.0	56	17
2	dark	0.0	56	14
3	light	7.2	80	28
4	dark	7.2	80	20
5	light	24.1	52	18
6	dark	24.1	52	22

Creating new variable: REMAIN

```
moth$REMAIN <- moth$PLACED - moth$REMOVED
```

	MORPH	DISTANCE	PLACED	REMOVED	REMAIN
1	light	0.0	56	17	39
2	dark	0.0	56	14	42
3	light	7.2	80	28	52
4	dark	7.2	80	20	60
5	light	24.1	52	18	34
6	dark	24.1	52	22	30

Did some morph have higher predation overall?

Call:

```
glm(formula = cbind(REMOVED, REMAIN) ~ MORPH, family = binomial,  
     data = moth)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-0.57752	0.09473	-6.097	1.08e-09 ***
MORPHlight	-0.40331	0.13925	-2.896	0.00377 **

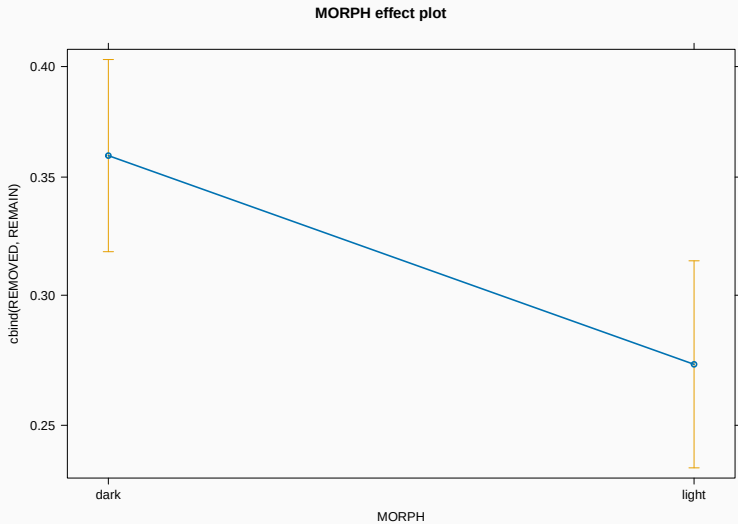
Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 35.385 on 13 degrees of freedom
Residual deviance: 26.936 on 12 degrees of freedom
AIC: 93.61

Number of Fisher Scoring iterations: 4

Did some morph have higher predation overall?



Did predation increase farther from city centre?

Call:

```
glm(formula = cbind(REMOVED, REMAIN) ~ DISTANCE, family = binomial,  
     data = moth)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-0.925861	0.136634	-6.776	1.23e-11 ***
DISTANCE	0.005268	0.003984	1.322	0.186

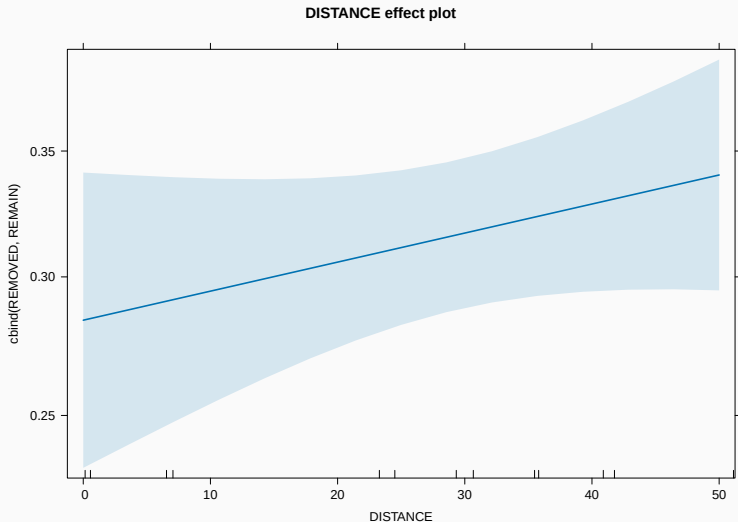
Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 35.385 on 13 degrees of freedom
Residual deviance: 33.626 on 12 degrees of freedom
AIC: 100.3

Number of Fisher Scoring iterations: 4

Did predation increase farther from city centre?



Did dark morph have lower predation in city & light have lower predation in countryside?

Call:

```
glm(formula = cbind(REMOVED, REMAIN) ~ MORPH * DISTANCE, family = binomial,  
     data = moth)
```

Coefficients:

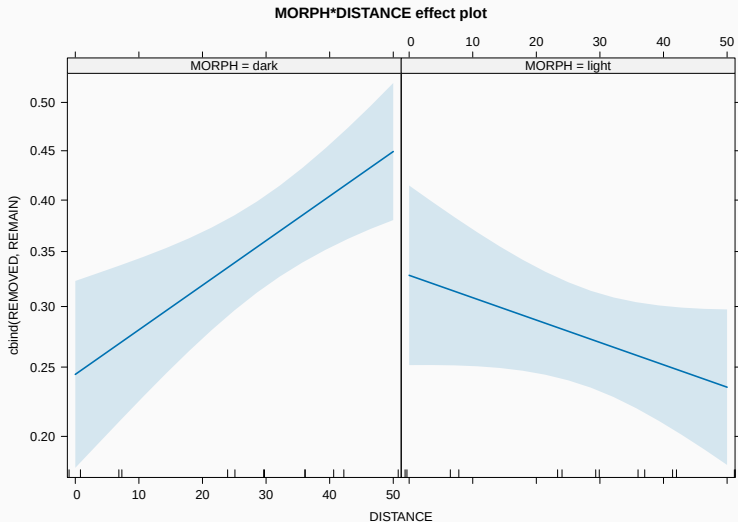
	Estimate	Std. Error	z value	Pr(> z)	
(Intercept)	-1.128987	0.197906	-5.705	1.17e-08	***
MORPHlight	0.411257	0.274490	1.498	0.134066	
DISTANCE	0.018502	0.005645	3.277	0.001048	**
MORPHlight:DISTANCE	-0.027789	0.008085	-3.437	0.000588	***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 35.385 on 13 degrees of freedom
Residual deviance: 13.230 on 10 degrees of freedom
AIC: 83.904

Did dark morph have lower predation in city & light have lower predation in countryside?



```
simulateResiduals(pred.int, plot = TRUE)
```

DHARMa residual

